

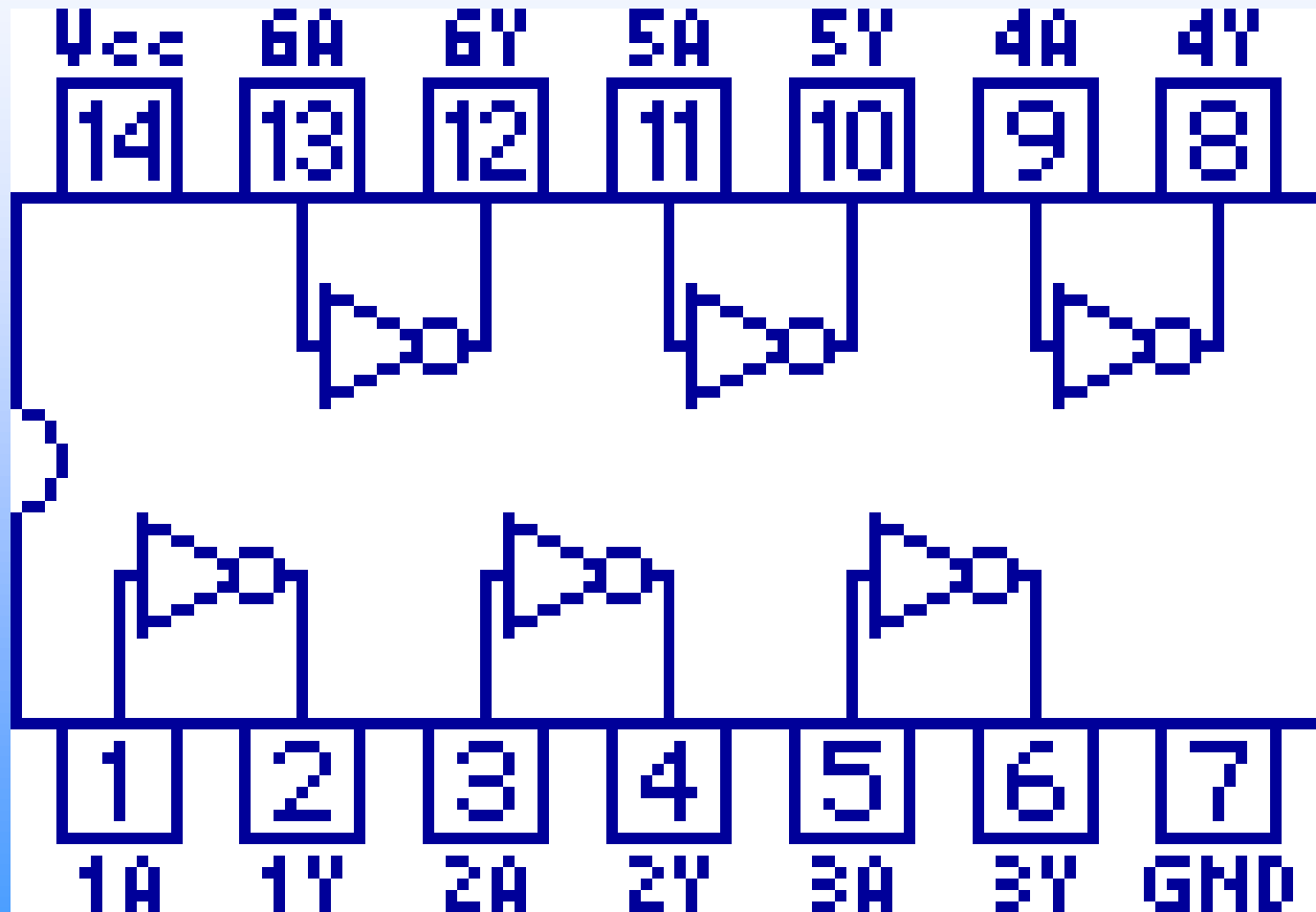
邏輯設計實驗

實驗1

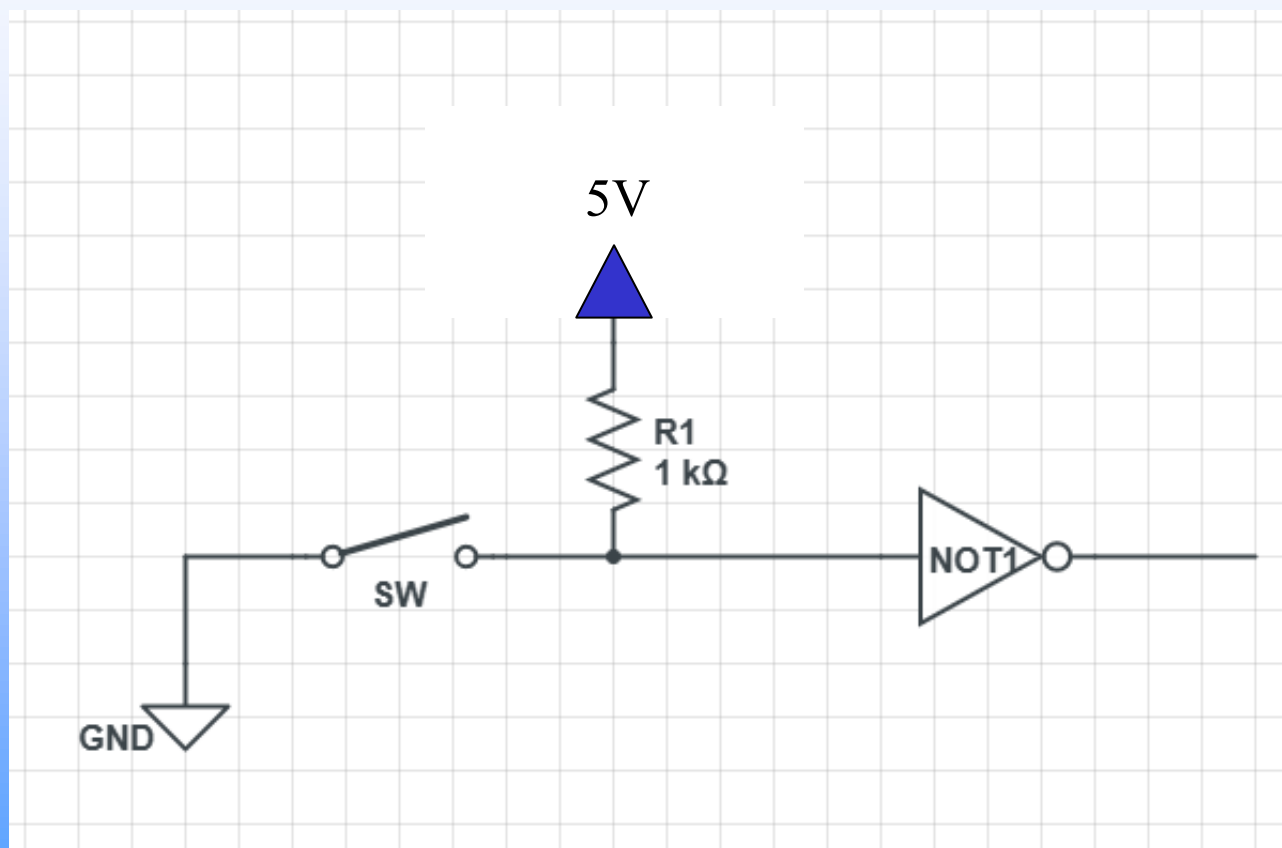
Q1

1. Use a DIP switch to adjust low or high as the input of a 7404 NOT gate.
 - a. Use a logic probe to test the input and the output of the gate.
 - b. Measure the input/output voltages of the gate when the input is low and high, respectively.
 - c. A green LED must be on when the output is high; otherwise, off.
 - d. A red LED must be on when the input is high.
 - e. Use a logic probe to test the input and the output of the gate.
 - f. Measure the input/output voltages of the gate when the input is low and high, respectively.
 - g. Compare the results of 1a and 1f.

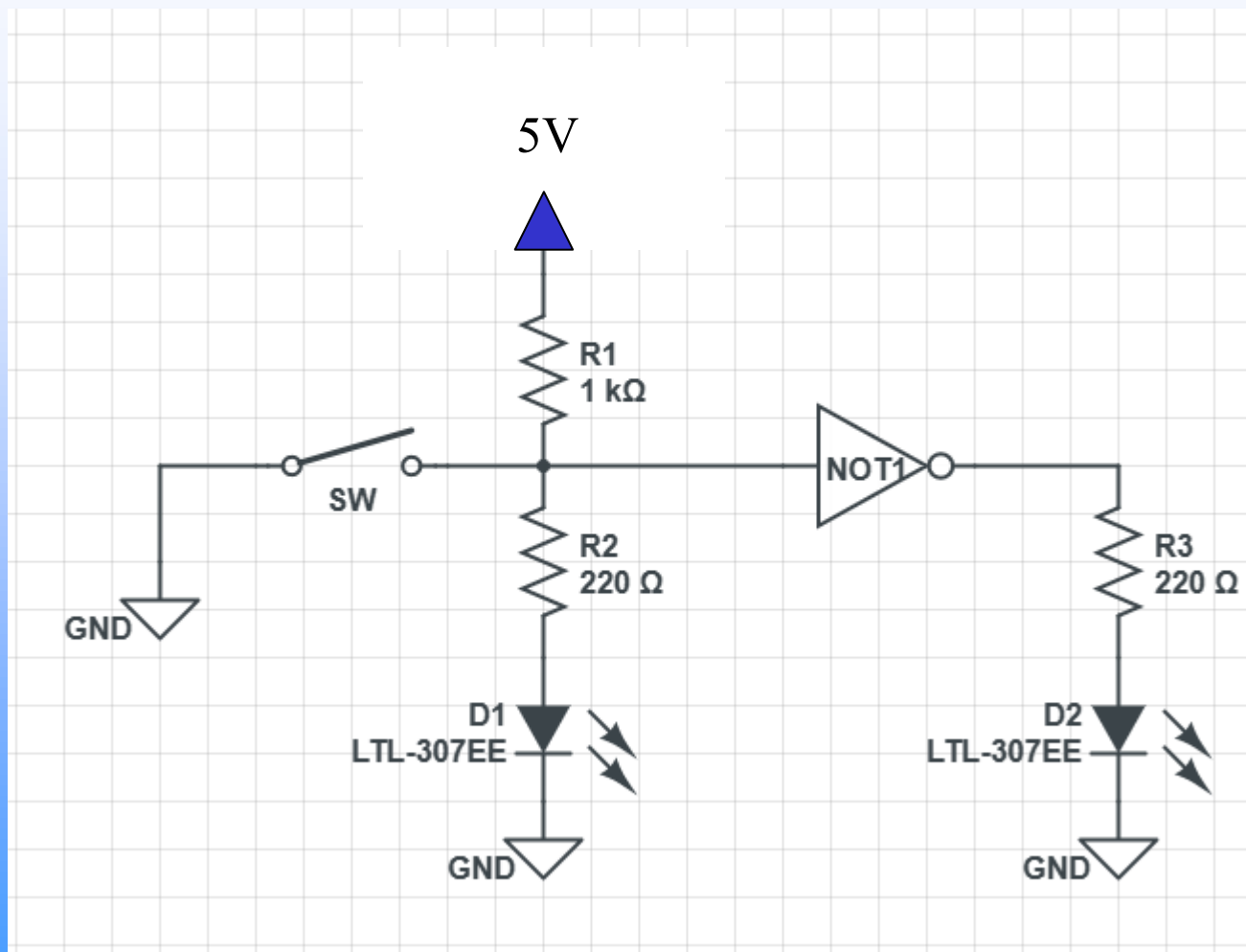
7404 NOT



電路圖(a,b)



電路圖 (c,d,e)



Q2

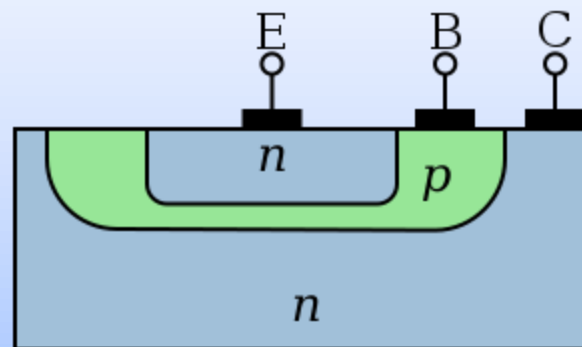
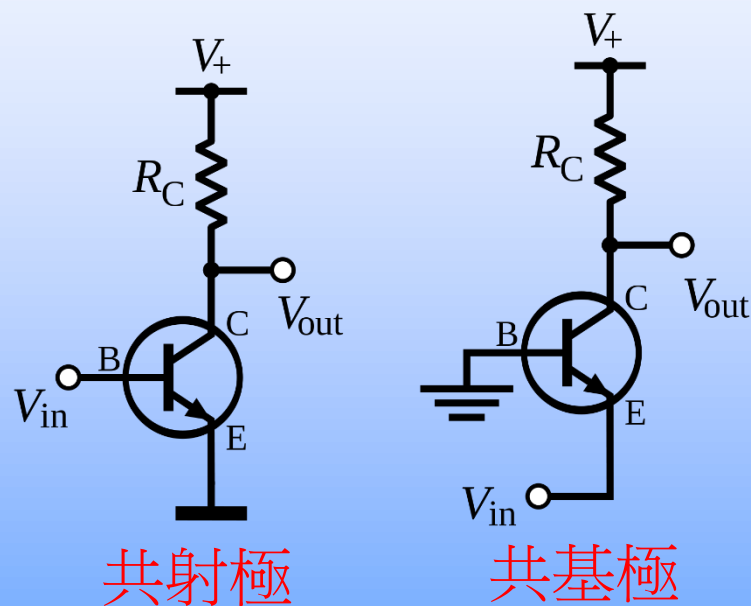
Use a DIP switch to adjust low or high as the input of a 7404 NOT gate.

A green LED must be on when the input of the gate is high.

Use the gate to drive an npn transistor. The transistor will be able to turn on a 3W-bulb when the output of the gate is high.

NPN (BJT)

BJT: **B**ipolar **J**unction **T**ransistor (雙極性電晶體)



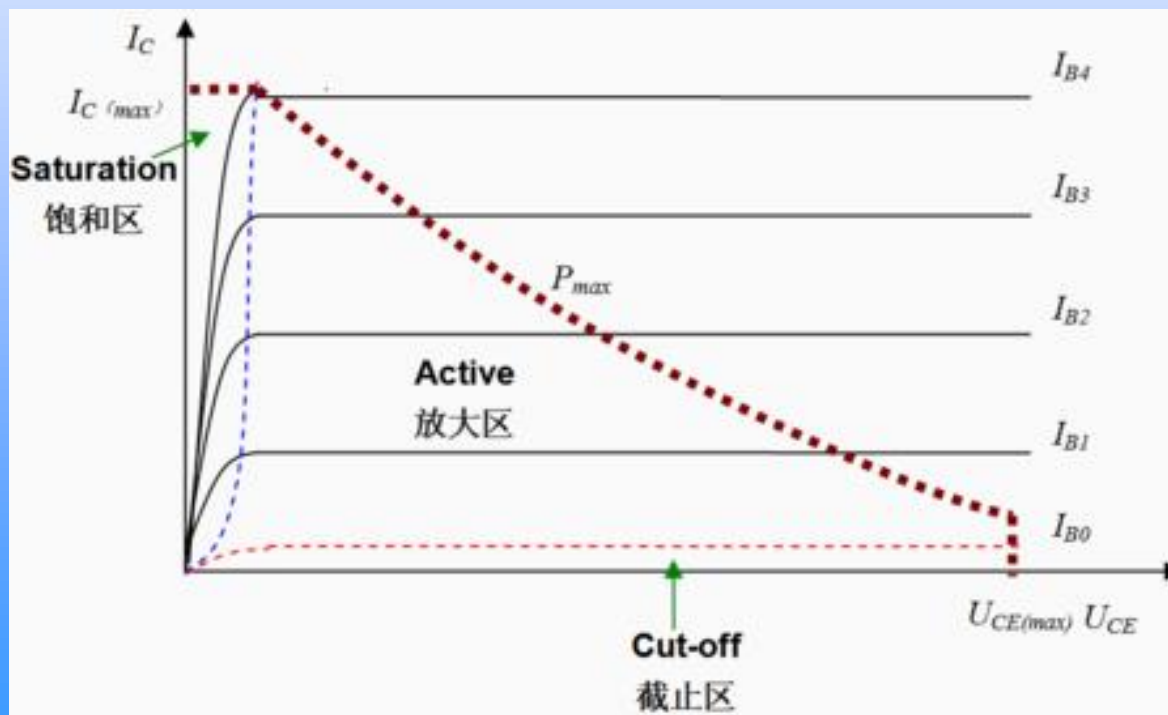
[Source: Bipolar junction transistor - Wikipedia](#)

[Source: 雙極性電晶體 - 維基百科，自由的百科全書 \(wikipedia.org\)](#)

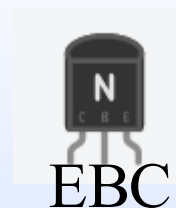
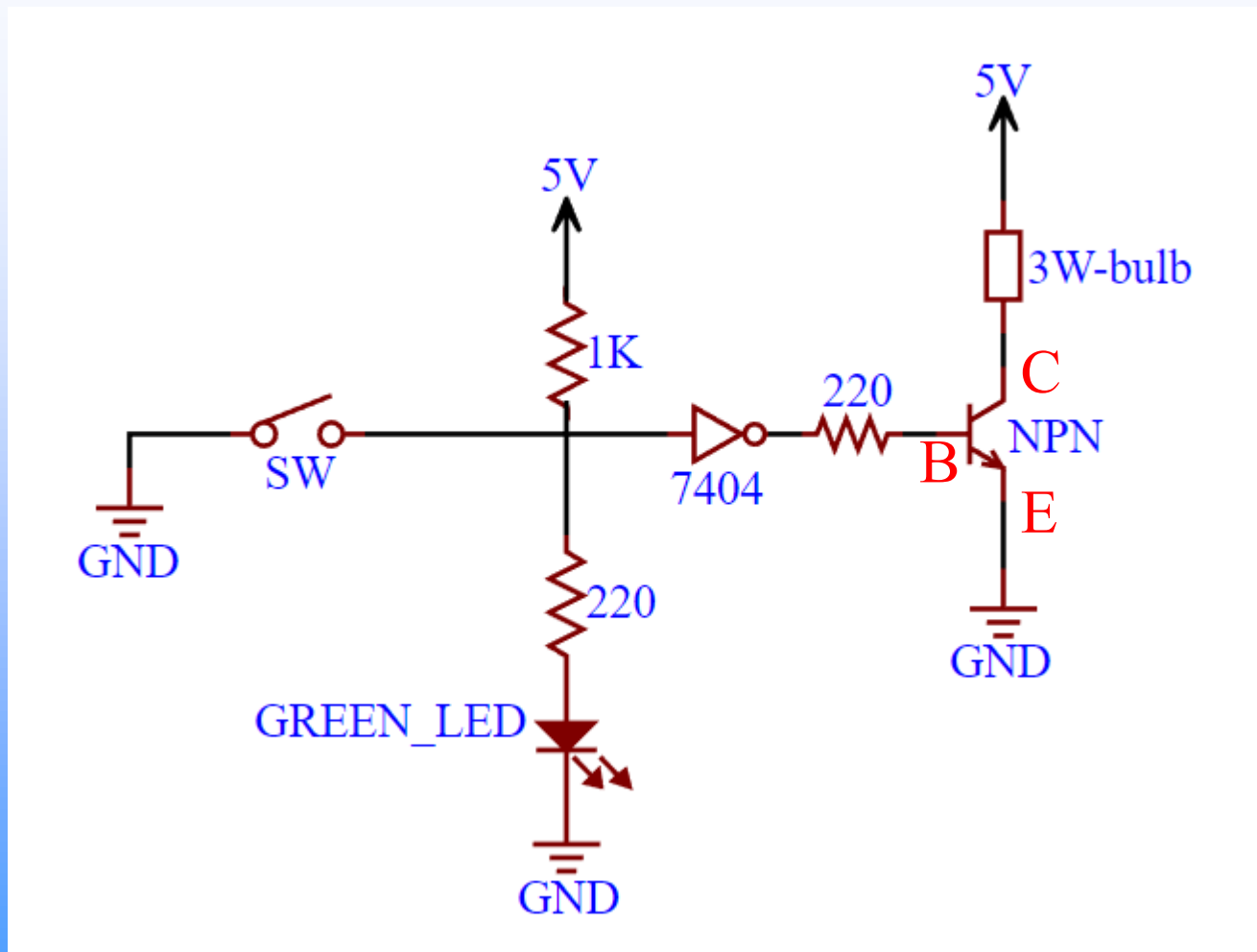
NPN (BJT)

BJT工作模式

模式	定義	應用
主動區	E順偏、C逆偏	放大訊號: 放大器、 濾波器、振盪器
飽和	E順偏、C順偏	開關、邏輯閘
截止	E、C皆逆偏	

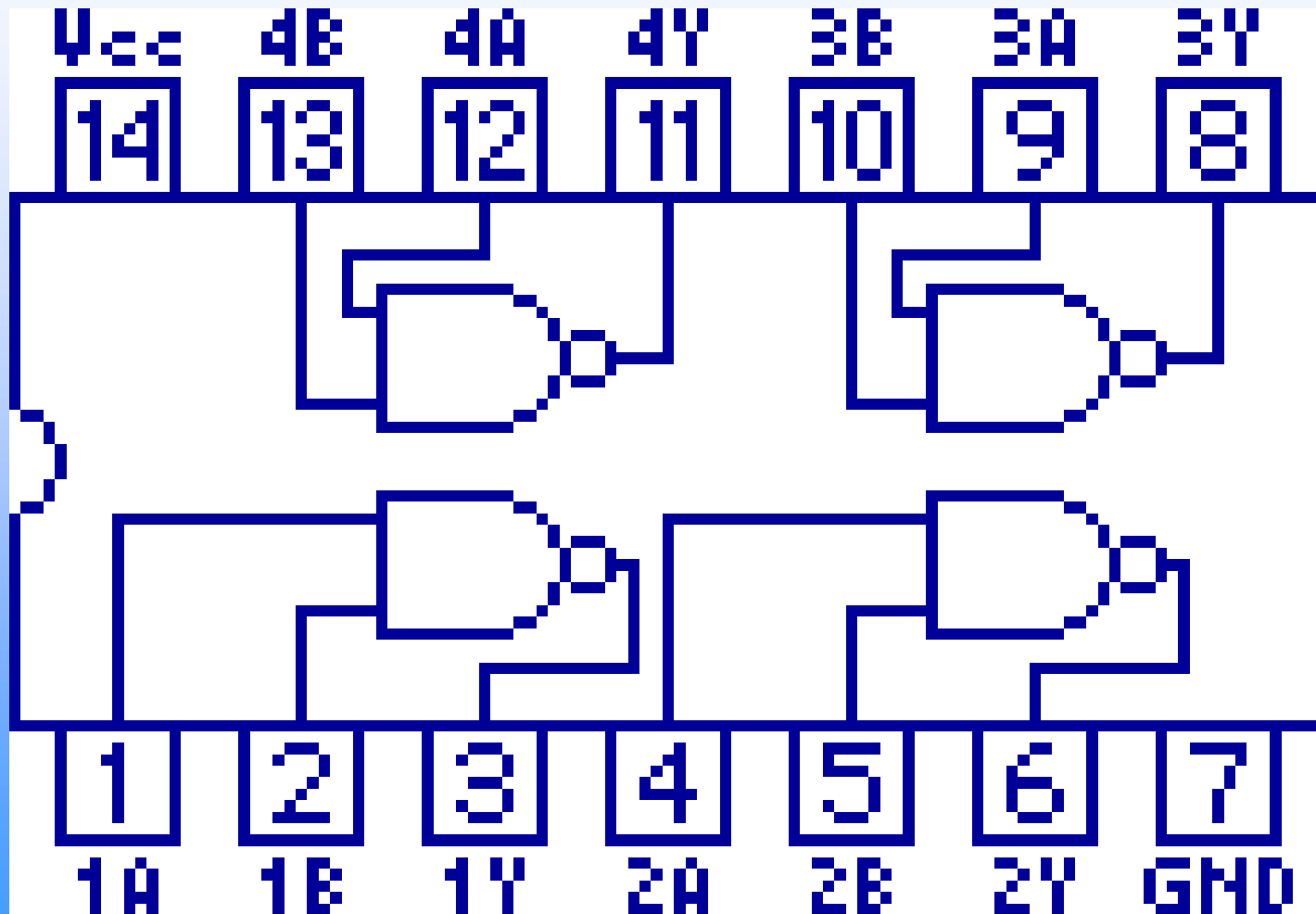


Q2



$$I_C = \beta I_B, I_E = (1 + \beta) I_B \quad \beta \text{ 在 } 50 \sim 1000 \text{ 之間}$$

7400 NAND

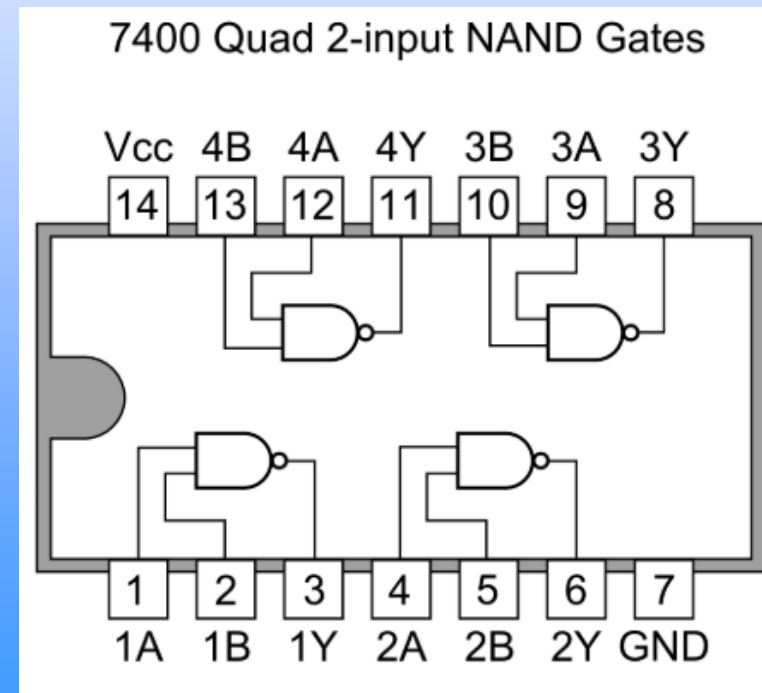
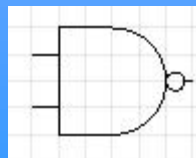


NAND Gate Experiment

1-3. Use DIP switches to adjust low or high as the inputs of a 7400 NAND gate, also use green LEDs to show high in inputs and a red LED to indicate low in output.

1-4. Measure the output voltage of a 7400 NAND gate without load when its output is high/low.

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0



NAND Gate Experiment

